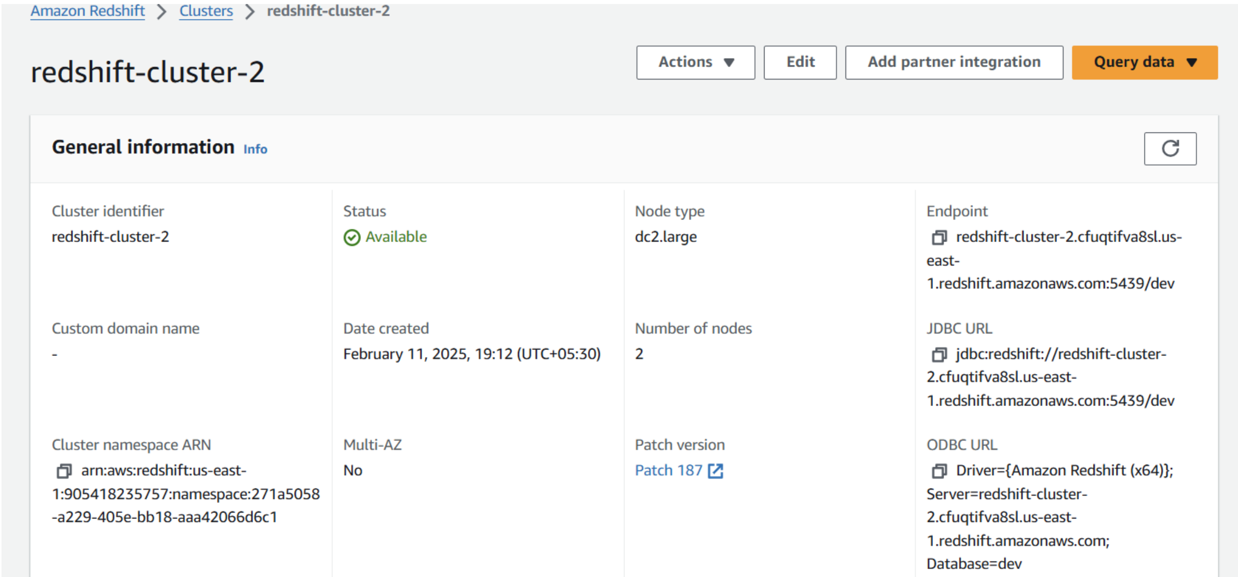
Creation of a Redshift Cluster

Screenshots of the configuration of the Redshift cluster that you have created:



Queries used for creating the Dimension and Fact tables on the Redshift cluster along with screenshots of the successful status of the query

Setting up a database in the Redshift cluster and running queries to create the dimension and fact tables

Queries to create the various dimension and fact tables with appropriate primary and foreign keys:

creating DIM_LOCATION table

create table if not exists etl_atm_data.DIM_LOCATION(
location_id int not null sortkey distkey,
location varchar(50),
streetname varchar(255),
street_number int,
zipcode int,

```
lat decimal(10,3),
lon decimal(10,3),
PRIMARY KEY(location_id)
);
```

creating DIM_ATM table

```
create table if not exists etl_atm_data.DIM_ATM(
atm_id int not null distkey sortkey,
atm_number varchar(20),
atm_manufacturer varchar(50),
atm_location_id int,
PRIMARY KEY(atm_id),
FOREIGN KEY(atm_location_id) REFERENCES
etl_atm_data.DIM_LOCATION(location_id)
);
```

creating DIM_DATE table

```
create table if not exists etl_atm_data.DIM_DATE(
date_id int not null distkey sortkey,
full_date_time timestamp,
year int,
month varchar(20),
day int,
hour int,
weekday varchar(20),
PRIMARY KEY(date_id)
);
```

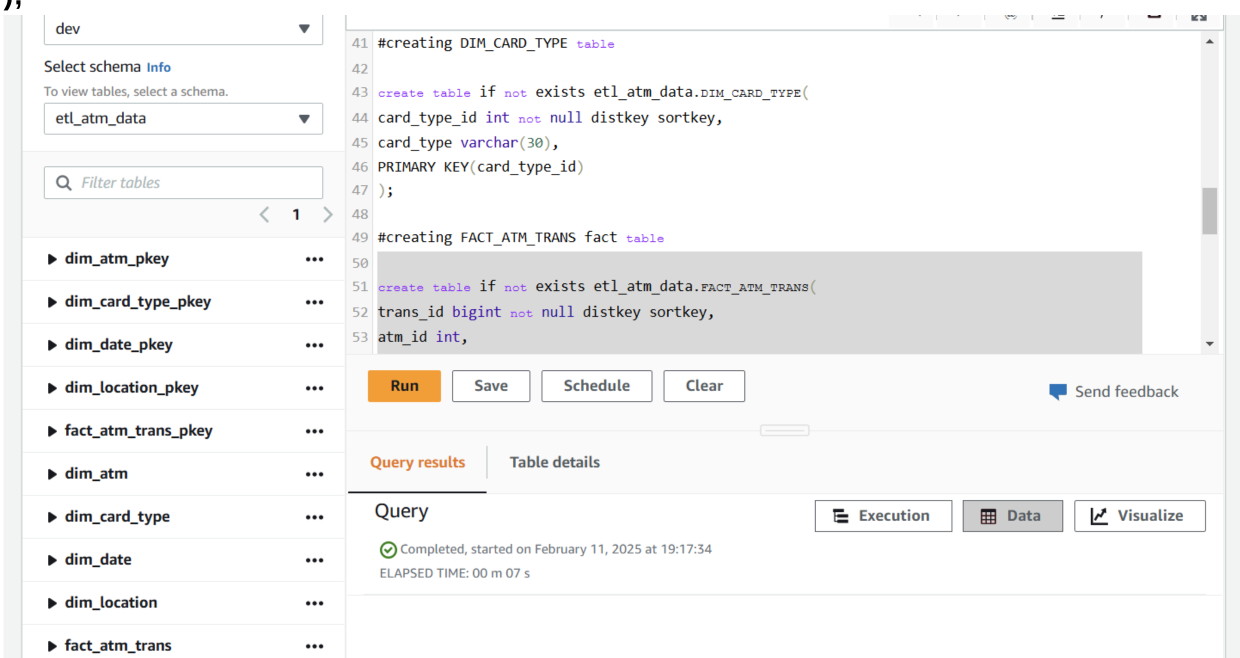
creating DIM_CARD_TYPE table

```
create table if not exists etl_atm_data.DIM_CARD_TYPE(
card_type_id int not null distkey sortkey,
card_type varchar(30),
PRIMARY KEY(card_type_id)
);
```

#creating FACT_ATM_TRANS fact table

```
create table if not exists etl_atm_data.FACT_ATM_TRANS(
trans_id bigint not null distkey sortkey,
```

```
atm_id int,
weather_loc_id int,
date_id int,
card_type_id int,
atm_status varchar(20),
currency varchar(10),
service varchar(20),
transaction_amount int,
message_code varchar(255),
message_text varchar(255),
rain_3h decimal(10,3),
clouds_all int,
weather_id int,
weather_main varchar(50),
weather_description varchar(255),
PRIMARY KEY(trans_id),
FOREIGN KEY(weather_loc_id) REFERENCES
etl_atm_data.DIM_LOCATION(location_id),
FOREIGN KEY(atm_id) REFERENCES etl_atm_data.DIM_ATM(atm_id),
FOREIGN KEY(date_id) REFERENCES etl_atm_data.DIM_DATE(date_id),
FOREIGN KEY(card_type_id) REFERENCES
etl_atm_data.DIM_CARD_TYPE(card_type_id)
);
```



Loading data into a Redshift cluster from Amazon S3 bucket

Queries to copy the data from S3 buckets to the Redshift cluster in the appropriate tables
#loading data into DIM_LOCATION table
copy etl_atm_data.DIM_LOCATION from 's3://aws-logs-905418235757-us-east-1/elasticmapreduce/j-1E6SEFG9GA4YQ/etl/dim_location/part-00000-4497a930-8930-45db-93a5-705c7d58d299-c000.csv' iam_role 'arn:aws:iam::905418235757:role/myRedshiftRole' delimiter ',' region 'us-east-1' CSV;

#loading data into DIM_ATM table
copy etl_atm_data.DIM_ATM from 's3://aws-logs-905418235757-us-east-1/elasticmapreduce/j-1E6SEFG9GA4YQ/etl/dim_atm/part-00000-dc0c784f-40a6-4330-a08d-553d23e030a5-c000.csv' iam_role 'arn:aws:iam::905418235757:role/myRedshiftRole' delimiter ',' region 'us-east-1' CSV;

#loading data into DIM_DATE table
copy etl_atm_data.DIM_DATE from ‘s3://aws-logs-905418235757-us-east-1/elasticmapreduce/j-1E6SEFG9GA4YQ/etl/dim_date/part-00000-339f7738-7b0f-47ba-8fed-c81f7c1579e2-c000.csv' iam_role 'arn:aws:iam::905418235757:role/myRedshiftRole' delimiter ',' region 'us-east-1' CSV timeformat
'auto';

#loading data into DIM_CARD_TYPE table
copy etl_atm_data.DIM_CARD_TYPE from 's3://aws-logs-905418235757-us-east-1/elasticmapreduce/j-1E6SEFG9GA4YQ/etl/dim_card_type/part-00000-666ce7dc-4bc4-4312-99db-c75d5c570b8f-c000.csv' iam_role 'arn:aws:iam::905418235757:role/myRedshiftRole' delimiter ',' region 'us-east-1' CSV;

#loading data into FACT_ATM_TRANS table
copy etl_atm_data.FACT_ATM_TRANS from 's3://aws-logs-905418235757-us-east-1/elasticmapreduce/j-1E6SEFG9GA4YQ/etl/fact_atm_trans/part-00000-f625ca03-24bb-4275-9a39-fb54a1eeb281-c000.csv' iam_role 'arn:aws:iam::905418235757:role/myRedshiftRole' delimiter ',' region 'us-east-1' CSV;